



The Path Is the Math

WisdomForge Booklet — Ages 5–10

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About This Book

This booklet is the six math sittings called *Show the Path*. Those sittings are **young+ only** — Little math already has *Number in the Age of AI*. Little sitting pages **404** by design. This elementary booklet still exists so a younger reader can hold the same law: a number you cannot walk is a rumor. The model may check. You walk.

It is not the *Number in the Age of AI* pack. It is not a little sitting unit. Do not invent sittings.

*“What is the unknown? What are the data? What is the condition?” — George Pólya,
How to Solve It (Princeton, 1945)*

Chapter 1: What Is a Proof?



Illustration for Chapter 1

A proof is not a fancy word. It is a path other people can walk, step by step, from what they know to what you are saying.

Objective. The student can explain what a proof is in plain language: a path of steps from something known to a conclusion, where each step is visible and walkable.

For the grown-up in the room. A proof is a path. That is the whole idea. You start somewhere everyone agrees on. You take a step. You show the step. You take another step. You show it. You arrive at the conclusion. If every step is visible and anyone could walk it, the proof holds. If a step is missing or invisible, the proof is broken. The model can produce the conclusion. It cannot always show the path. The student needs to learn that the path is the proof, not the answer.

The path. Imagine you tell a friend your house is two blocks from school. They say: prove it. You do not just say 'it is.' You walk the path. Left on Oak. Right on Maple. Two blocks. Here is the door. That is a proof. You showed each step. Your friend could walk it. A mathematical proof is the same thing with ideas instead of streets. You start somewhere everyone agrees. You take a step. You show it. You take another. You show it. You arrive. The path is the proof. The answer is the door. Without the path, the answer is just a claim.

The model's shortcut. A model will give you the answer. 'The sum is 42.' It will not always show the path. It jumps to the door. That is useful if you only need the answer. It is not a proof. A proof is the path. If you cannot walk the path yourself, you have the answer but not the proof. The model is a shortcut machine. Shortcuts are useful. They are not mathematics. Mathematics is the path. The student who confuses the shortcut for the path will have answers and no understanding. The student who knows the difference can use the shortcut when the path does not matter and walk the path when it does.

Why the path matters. The answer to a math problem is a destination. The path is how you get there. If someone gives you the destination, you know where it is. You do not know how to get there. If you walk the path, you know both. The next problem is a different destination, but the path uses the same streets. The student who walks paths learns the streets. The student who collects answers learns nothing reusable. The model gives destinations. It is a taxi. Taxis are useful. They do not teach you to walk. Walk the path yourself. Use the taxi when you already know the streets.

Using the model as a path-checker. The model can check your path. You walk the steps. You show the model your steps. You ask: 'Is each step legal? Did I skip one?' The model reviews the path. It does not replace the walking. If the model says 'your step 3 is wrong,' you fix it. If the model says 'this is correct,' you still verify. The model is a checker, not a walker. The student who uses it this way gets the best of both: their own understanding and a tireless reviewer. The student who lets the model walk the path gets neither.

Sitting big idea. A proof is a path other people can walk. The model gives the door. The proof is the walk.

If they say “The answer is what matters.”: The answer matters for the task. The path matters for understanding. If you only have the answer, you cannot do the next problem. If you have the path, you can walk it again.

If they say “The model's path is clearer than mine.”: It is. That is because it has walked a million paths. Clarity is not understanding. Walk your own path, even if it is messier. The mess is where you learn. The model's clean path is a demonstration, not your learning.

If they say “I only need the answer for the test.”: For the test, the answer might be enough. For the next course, the next problem, the next job, it is not. The path is reusable. The answer is disposable. Choose the reusable one.

Integrity. Do not copy the model's answer and call it your proof. The proof is the path. If you did not walk it, it is not yours.

Dinner. What path did we walk today, and could someone else follow it?

- Do not use the word 'theorem' yet. Use 'path.' The vocabulary comes later.
- Keep it to simple examples: arithmetic steps, a geometric argument, a logic puzzle.

Words to Remember

*“What is the unknown? What are the data? What is the condition?” — George Pólya, *How to Solve It* (Princeton, 1945)*

*“A proof is a path other people can walk. The model gives the door. The proof is the walk.” — WisdomForge sitting *What Is a Proof?**

Big Idea

The path is the mathematics. A boxed number without steps is a prize, not a proof. The model may check. You walk.

Try This

1. One problem. Write the steps on paper. No tool first.
2. Point to each step. Is any missing?
3. Ask a grown-up to walk your path. Then a tool may check — not walk for you.

Ask a Grown-Up

What path did we walk today, and could someone else follow it?

Chapter 2: The Missing Step



Illustration for Chapter 2

A proof with a hole in it is not a proof. It is a guess with a gap.

Objective. The student can identify a missing step in a simple mathematical argument and explain why the gap breaks the path.

For the grown-up in the room. A proof is a path. If a step is missing, the path has a hole. You cannot walk across a hole. The model is very good at jumping over holes. It produces the answer without showing the step that connects. The student needs to learn to look for holes. If the model says 'and then the answer is 42,' the student asks: 'What step got you from the line before to 42?' If there is no step, there is a hole. The hole is the lesson.

The hole in the path. A path has steps. If you remove a step, the path has a hole. You can see the other side, but you cannot walk there. The model is very good at jumping. It says 'the answer is 42' and skips the step that gets you there. If you cannot see the hole, you will trust the jump. If you can see it, you will ask: 'Where did 42 come from?' That question is the skill. The model's jump is not a proof. It is a leap. A proof is a walk, not a leap.

Finding the hole. To find the hole, read each step and ask: 'Does this step follow from the one before it?' If yes, walk on. If you cannot see how it follows, there is a hole. The hole might be a missing calculation, a skipped assumption, or a logical jump. Mark the hole. Ask: 'What step would fill this gap?' If you can fill it, the path holds. If you cannot, the path is broken. The model will not mark its own holes. You have to find them. That is the student's job, not the model's.

The gap audit. A gap audit is simple. Read each step of the model's solution. For each step, ask two questions: 'Does this follow from the previous step?' and 'Can I reproduce this step on a blank board?' If both answers are yes, the step is real. If either is no, there is a gap. Write GAP. Then decide: can I fill it? If you can fill it with your own work, the path holds, and you have made the step yours. If you cannot fill it, the path is broken at that point. Do not trust the solution past the gap. The gap is not a minor inconvenience. It is a structural failure in the proof.

The invisible gap. The most dangerous gap is the one that looks like a step. The model writes 'therefore' or 'so' and jumps. The jump looks like a step because it has the right words around it. But the words 'therefore' and 'so' do not create a logical connection. They announce one. If the connection is not shown, the step is a gap dressed as a step. The student who reads 'therefore' and trusts it has been fooled by the costume. The student who reads 'therefore' and asks 'what follows from what?' has started the audit. The audit is the skill. The model will not perform it for you.

Sitting big idea. A proof with a missing step is a guess with a gap. The model jumps. The student finds the hole.

If they say "The model probably skipped it because it's obvious.": Obvious to whom? If it is obvious to you, fill it. If it is not obvious to you, it is a hole. 'Obvious' is the most common word used to hide a gap. Check it.

If they say “The model's solution looked fine to me.”: It looked fine because the gaps were dressed as steps. The audit is how you see the costume. If you did not find gaps, either the path was complete or you missed them. Audit again with a harder problem.

If they say “I can't fill the gap, but the answer is probably right.”: 'Probably right' is not a proof. If you cannot fill the gap, you have an unsupported step. The answer might be right. The proof is broken. Those are different things. A right answer with a broken proof is a coincidence, not a proof.

Integrity. If you find a hole, say so. Do not pretend the path was complete because the answer was right. A right answer with a hole is still a broken proof.

Dinner. What hole did we find in a path this week, and did we fill it?

- Keep the examples simple: arithmetic with a skipped step, a logic puzzle with a jump.
- Do not shame the model for skipping. The point is the student's ability to see the gap.

Words to Remember

*“What is the unknown? What are the data? What is the condition?” — George Pólya, *How to Solve It* (Princeton, 1945)*

*“A proof with a missing step is a guess with a gap. The model jumps. The student finds the hole.” — WisdomForge sitting *The Missing Step**

Big Idea

The path is the mathematics. A boxed number without steps is a prize, not a proof. The model may check. You walk.

Try This

1. One problem. Write the steps on paper. No tool first. 2. Point to each step. Is any missing?
3. Ask a grown-up to walk your path. Then a tool may check — not walk for you.

Ask a Grown-Up

What hole did we find in a path this week, and did we fill it?

Chapter 3: The Model Checks, You Walk



Illustration for Chapter 3

You walk the path. The model checks your steps. That is the right order. Flip it and you lose the path.

Objective. The student can solve a problem on their own, then ask the model to check each step, and explain why walking first and checking second is the right order.

For the grown-up in the room. This sitting establishes the order: walk first, check second. The student solves the problem. Then the model reviews. If the model goes first, the student copies the path and calls it their own. If the student goes first, the model is a checker, not a walker. The order is the lesson. Walk first. Check second. The model is a reviewer, not a replacement.

The right order. You walk the path first. You write each step. You arrive at the answer. Then you ask the model: 'Check my steps.' The model reviews. It finds a gap. You fix it. It confirms a step. You keep it. The model is a checker. You are the walker. If you flip the order, the model walks and you copy. The copy is not a proof. It is a transcript of someone else's walk. The order is the lesson: walk first, check second. The model is a reviewer, not a replacement.

When you are stuck. Sometimes you cannot walk. You are stuck at a step. You may ask the model for a hint: 'What kind of step should come next?' Not: 'What is the answer?' A hint is a nudge. It points to the street. You still walk it. A solution is a taxi. It takes you to the door. You did not walk. The student who asks for hints learns to walk. The student who asks for solutions learns to ride. Walk. Ask for hints when stuck. Never ask for the solution.

The hostile review. In professional mathematics, proofs are reviewed. The reviewer does not rewrite the proof. They attack it. They find the weak step, the unjustified claim, the missing assumption. The author fixes it. The proof gets stronger. The model can be that reviewer. You walk the path. You show the model each step. You ask: 'Attack this. Where is the weakest step? Where would a skeptic push?' The model finds the weak points. You fix them. The model does not rewrite. You rewrite. The model is the reviewer. You are the author. That is the right relationship.

Why hostile, not friendly. A friendly model says 'your proof looks good.' That is useless. A hostile model says 'step 3 assumes something you did not prove. Step 5 jumps without justification.' That is useful. The student who asks for a friendly review gets nothing. The student who asks for a hostile review gets the weak points. Ask the model to be hostile: 'Find the weakest step. Tell me where a skeptic would push.' The model is better at being hostile than at being right. Use that. The hostile review is the value. The friendly review is flattery.

Sitting big idea. Walk first. Check second. The model is a reviewer, not a replacement. When stuck, ask for hints, not answers.

If they say “It's faster to just ask the model.”: It is faster. Faster is not the skill. Walking is the skill. The model can check your walk. It cannot walk for you and call it yours. Walk first.

If they say “The model said my proof was good.”: Ask it to be hostile. 'Find the weakest step.' A model that says 'good' is being friendly, not useful. A model that finds the weak step is being a reviewer. Demand the review, not the compliment.

If they say “I'll just let the model fix it.”: The model finds the weakness. You fix it. If the model fixes it, the fix is the model's, not yours. The proof is only yours if you wrote every fix. The model is the reviewer. You are the author.

Integrity. If the model filled a gap for you, cite it. The gap is the model's step, not yours. The proof is only yours if you walked every step.

Dinner. What path did we walk first this week, and did the model check or replace?

- If the student is stuck, they may ask the model for a hint, not a solution. A hint is a nudge. A solution is a replacement.

- Do not let the student copy the model's path and call it their proof. The walk is the proof.

Words to Remember

*“What is the unknown? What are the data? What is the condition?” — George Pólya, *How to Solve It* (Princeton, 1945)*

*“Walk first. Check second. The model is a reviewer, not a replacement. When stuck, ask for hints, not answers.” — WisdomForge sitting *The Model Checks, You Walk**

Big Idea

The path is the mathematics. A boxed number without steps is a prize, not a proof. The model may check. You walk.

Try This

1. One problem. Write the steps on paper. No tool first.
2. Point to each step. Is any missing?
3. Ask a grown-up to walk your path. Then a tool may check — not walk for you.

Ask a Grown-Up

What path did we walk first this week, and did the model check or replace?

Chapter 4: The Blank Board Test



Illustration for Chapter 4

If you close the book and cannot do it again, you did not learn it. You read it.

Objective. The student can reproduce one key step from a problem they solved with the model, without the model or the notes, on a blank board.

For the grown-up in the room. This is the reproducibility test. The student solves a problem with the model's help. Then they close everything. Blank board. Can they reproduce the key step? If yes, they learned it. If no, they read it. The model is a reading machine. Reading is not learning. The blank board test is the check. Use it once on one problem. The child who passes has learned. The child who fails has read. Both are useful data.

Reading is not learning. You can follow a model's steps while it shows them. That is reading. You close the chat. Can you do it again? If yes, you learned. If no, you read. Reading is useful. It is not the same as learning. Learning is what you can reproduce on a blank board. The blank board test is simple: close everything, take a blank page, reproduce the key step. If you can, the path is yours. If you cannot, the path was the model's. You followed it. You did not own it. The blank board test is the difference between following and owning.

The standard. In mathematics, reproducibility is the standard. A proof is yours if you can reproduce it. A proof is the model's if you cannot. The blank board test is the check: close everything, take a blank page, reproduce the proof. If you can, the path is in your hands. If you cannot, the path was in the model's hands and you were reading over its shoulder. Both are useful experiences. Only one is understanding. The student who adopts reproducibility as the standard will always know the difference. The student who does not will confuse following for owning for the rest of their education.

What the test reveals. The blank board test reveals exactly which steps you own and which you followed. You might own steps 1 through 3 and lose the path at step 4. That is useful data. Step 4 is the step you need to relearn. The model can help you relearn it. Then you take the blank board test again. If you can reproduce step 4 this time, you own it now. The test is not a grade. It is a diagnostic. It tells you where the path is yours and where it is not. Then you fix the gaps. The test is the map of your understanding.

Sitting big idea. If you cannot reproduce it on a blank board, you read it. You did not learn it. The blank board test is the check.

If they say "I understood it while the model was showing it.": Understanding while reading is real. It is not the same as reproducing. The blank board test tells you which one you have. If you can reproduce it, you own it. If you cannot, you understood it while it was in front of you. Both are useful. Only one is learning.

If they say "I don't need to reproduce it. I understood it.": Understanding while following is real. It is not the same as reproducing. The blank board test is the difference. If you can reproduce it, you own it. If you cannot, you understood it while it was in front of you. That is a different thing. Both are useful. Only one is mastery.

If they say “In the real world, I can just look it up.”: You can look up the steps. You cannot look up the understanding. The steps are on the internet. The path in your hands is not. The blank board test is how you know which one you have.

Integrity. Do not peek at the model's solution during the blank board test. If you peek, the test is invalid. The peek is the data: you could not reproduce it.

Dinner. What could we reproduce on a blank board today, and what could we not?

- Do not shame the child for failing the blank board test. The failure is data, not a grade.
- Keep it to one key step, not the whole problem. The point is the test, not the volume.

Words to Remember

*“What is the unknown? What are the data? What is the condition?” — George Pólya, *How to Solve It* (Princeton, 1945)*

*“If you cannot reproduce it on a blank board, you read it. You did not learn it. The blank board test is the check.” — WisdomForge sitting *The Blank Board Test**

Big Idea

The path is the mathematics. A boxed number without steps is a prize, not a proof. The model may check. You walk.

Try This

1. One problem. Write the steps on paper. No tool first. 2. Point to each step. Is any missing?
3. Ask a grown-up to walk your path. Then a tool may check — not walk for you.

Ask a Grown-Up

What could we reproduce on a blank board today, and what could we not?

Chapter 5: A Proof Is for Other People



Illustration for Chapter 5

A proof is not for you. It is for the person who needs to walk it after you. Write it so they can.

Objective. The student can write a simple proof in steps that another person could follow without help, and explain why a proof is for the reader, not the writer.

For the grown-up in the room. A proof is a social act. It is not for you. It is for the person who reads it. If only you can follow it, it is a note, not a proof. A proof is written so that someone else can walk the path. The model writes proofs for itself. It jumps. It skips. It assumes. The student needs to learn to write for the reader. The reader is the test: could a friend walk your steps? If yes, it is a proof. If no, it is a note.

The reader is the test. A proof is a path. The path is for the person who walks it after you. If you write the proof for yourself, you will skip steps that are obvious to you. The reader will fall in the gap. If you write for the reader, you show every step, even the ones that feel obvious. The test is simple: could a friend who has not seen this problem walk your steps and arrive at your answer? If yes, it is a proof. If no, it is a note to yourself. Both are useful. Only one is a proof.

The model writes for itself. The model writes proofs that skip steps because the steps are obvious to the model's pattern. They are not obvious to you. The model's proof is a note to itself, not a proof for you. When you read it and cannot follow, the gap is not your failure. It is the model's choice to write for itself, not for you. The skill is to read the model's proof and fill the gaps it left for itself. Then write your own proof, for your reader, with every step shown.

Mathematics is social. Mathematics is not done in private. A proof is written for a community. The community verifies it. If the community cannot follow it, the proof is not accepted. This is not a school rule. It is how mathematics works. The model writes proofs that skip steps because it does not have a reader. It has a pattern. The student who writes for a reader is doing mathematics. The student who writes for themselves is keeping notes. The difference is the reader. Write for the reader. Use the model as a test reader. Use a peer as a real reader. The proof that survives both is a proof.

The model as a test reader. The model can be a test reader. You write the proof. You show the model. You ask: 'Could a peer follow this? Where would they get stuck?' The model finds the gaps. You fill them. This is different from asking the model to write the proof. The model as a reader tests your proof. The model as a writer replaces your proof. Use it as a reader. Never use it as a writer. The student who knows the difference is doing mathematics with a tool. The student who does not is outsourcing their thinking.

Sitting big idea. A proof is for the reader, not the writer. Write it so a friend could walk it. The model writes for itself. You write for others.

If they say “It’s obvious to me.”: It is obvious to you. Is it obvious to the reader? The proof is for the reader. If the reader cannot follow, the proof is a note, not a proof. Fill the step.

If they say “I don’t need to write for others. I understand it.”: You understand it for yourself. That is a note. A proof is for the reader. If you cannot write it so another person can follow, you have a private understanding, not a public proof. Mathematics is public. Write for the reader.

If they say “The model can just write the proof.”: It can. That proof is the model’s, not yours. It is written for the model’s pattern, not for your reader. Use the model as a test reader. Write the proof yourself. The proof that is yours is the one you wrote for a reader.

Integrity. Do not present the model’s proof as your own. If the model wrote the steps and you filled the gaps, cite both. The proof is yours if you wrote it for the reader.

Dinner. What proof did we write for a reader this week, and could they walk it?

- Keep it to simple proofs: arithmetic, a basic geometric argument.
- Do not grade the child’s proof style. Grade whether another person could follow it.

Words to Remember

*“What is the unknown? What are the data? What is the condition?” — George Pólya, *How to Solve It* (Princeton, 1945)*

*“A proof is for the reader, not the writer. Write it so a friend could walk it. The model writes for itself. You write for others.” — WisdomForge sitting *A Proof Is for Other People**

Big Idea

The path is the mathematics. A boxed number without steps is a prize, not a proof. The model may check. You walk.

Try This

1. One problem. Write the steps on paper. No tool first. 2. Point to each step. Is any missing?
3. Ask a grown-up to walk your path. Then a tool may check — not walk for you.

Ask a Grown-Up

What proof did we write for a reader this week, and could they walk it?

Chapter 6: Show the Path



Illustration for Chapter 6

You now know: the path is the proof, the model checks, the blank board tests, and the path is for others. Show one full path.

Objective. The student can walk one full mathematical path: solve, write steps for a reader, check with the model, reproduce on a blank board, and report.

For the grown-up in the room. This is the capstone. The student runs the full cycle: solve a problem, write the steps for a reader, check with the model, reproduce on a blank board, report. The cycle ties together every skill from the unit. The output is one path, shown end to end, that a reader could walk and the student could reproduce. That is the capstone artifact.

The whole unit in one path. Walk a path. Write it for a reader. Check it with the model. Reproduce it on a blank board. Report. That is the whole unit in one cycle. The path is the proof. The model is the checker. The blank board is the test. The reader is the social. The report is the honesty. The child who can run this cycle once has learned the unit. The child who runs it again has a habit. The child who runs it for life has a skill that the model cannot replace: the ability to show a path others can walk.

The three-test proof. The capstone proof passes three tests. The reader test: a peer can walk it. The model's hostile review: the model found the weak steps and the student fixed them. The blank board test: the student can reproduce it without notes. A proof that passes all three is a proof. A proof that passes one or two is a draft. The student who runs the cycle and passes all three has produced something the model cannot: a path that is owned, verified, and reproducible. That is mathematics in the age of fluency.

Sitting big idea. Walk, write for a reader, check, reproduce, report. The whole unit in one path. That is the capstone.

If they say "That's a lot for one problem.": The first time, it is. The second time, it is faster. The third time, it is a habit. The cycle is an investment. The return is a path you own, that a reader can walk, and that the model checked. That is worth twenty minutes.

If they say "This is too much work for one proof.": The first time, it is. The capstone is the investment. The second proof will be faster. The third will be a habit. The capstone is the proof that the skill exists. After that, the skill is yours.

If they say "The model can write a better proof.": It can write a smoother proof. It cannot write a proof that passes your blank board test. It cannot write a proof that a peer walks. It can write a proof that a reader follows, but the path is the model's, not yours. The capstone is the proof that is yours.

Integrity. The report includes the path, the model's check, and the blank board result. Nothing is erased. The full cycle is the honest report.

Dinner. What full path did we show this week, and what did the cycle reveal?

- The student must run every step. Skipping a step is skipping the capstone.
- Celebrate the cycle, not the correctness. A wrong answer with a complete path is more valuable than a right answer with no path.

Words to Remember

*“What is the unknown? What are the data? What is the condition?” — George Pólya, *How to Solve It* (Princeton, 1945)*

*“Walk, write for a reader, check, reproduce, report. The whole unit in one path. That is the capstone.” — WisdomForge sitting *Show the Path**

Big Idea

The path is the mathematics. A boxed number without steps is a prize, not a proof. The model may check. You walk.

Try This

1. One problem. Write the steps on paper. No tool first.
2. Point to each step. Is any missing?
3. Ask a grown-up to walk your path. Then a tool may check — not walk for you.

Ask a Grown-Up

What full path did we show this week, and what did the cycle reveal?

For the Grown-Up Reader

Young+ sittings only. Little 404 by design. Pólya, *How to Solve It*, 1945. *Number in the Age of AI* is a different pack. Photomath/CAS in scope as checkers, not walkers. Cite the tool if it changed a step.

About WisdomForge

This is the elementary band of *Show the Path*. Payment pulled. No purchase CTA.